

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built a **distributed executor** routing pipeline nodes to **Docker** or **AWS Lambda** via typed **REST APIs**, backed by an **S3** content-addressed **artifact store** with memoization, sustaining **sub-200ms p99 latency** at scale.
- Built fault-tolerant **GPU/HPC scheduling** across **AWS and GCP storage and compute**, maintaining fleet utilization through **node loss and provider hiccups** with **Go and Python** placement and resource-allocation logic.
- Provisioned and operated **NVIDIA GPU clusters** supporting long-running **distributed training** and high-availability inference across a shared, **large-scale** fleet.
- Built a deployment-gated **evaluation framework** with **guardrails**, owning **design reviews** and **production debugging** and participating in **code reviews**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by rebuilding synchronous **AI** serving into an **async**, **autoscaled** architecture on **GCP Cloud Run**.
- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** and **re-ranking**.
- Designed a **Neo4j** knowledge graph for **graph-aware retrieval** in production, modeling domain entities and relationships.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Operated **25+ Java and Python microservices** serving **1M+ users** at **99.9% uptime** over 4 years on **AWS (EC2, RDS, SQS)**, owning **system design and operations**.
- Cut **production escalations 60%** in one quarter by standardizing **retry, timeout, and error-propagation** contracts across **distributed services**, and drove **root cause analysis** and **post-incident reviews**.
- Cut a nightly SQL aggregation pipeline from **8 minutes to 90 seconds** by moving to **indexed views** and **batch partitioning**, eliminating row-level lock contention.
- Led **architecture and code reviews** across production services, evaluating scalability, reliability, testability, and operational tradeoffs while mentoring engineers on **system design** and **reliability**.

PROJECTS

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Built a self-hosted **LLM observability** platform that traces prompts, responses, tokens, **latency**, and cost through a Python SDK decorator, sending traces asynchronously to a **FastAPI** collector on **PostgreSQL**, with **hallucination scoring** and a React dashboard.

LangFetch

tinyurl.com/LangFetch

- Built an **agentic SQL workspace** that plans multi-step queries, runs them in sequence, and reads each result before choosing the next, served via **FastAPI** with **Claude** and exposed as an **MCP** server.

VPC Control Plane Simulator

github.com/ganeshhgupta/vpc-control-plane-sim

- Built an AWS-style **VPC networking** simulator with **CIDR validation** and overlap detection, **longest-prefix-match routing**, stateful **security group** and stateless **NACL** evaluation, and **drift detection**, on **Next.js** with **PostgreSQL**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Go, Python, C++, Java, SQL

Infrastructure: Distributed Systems, Distributed Scheduling, Fault-Tolerant Placement, Storage, Docker, AWS, GCP

Software Engineering: Systems Architecture, Design Reviews, Code Review, Production Debugging, Technical Leadership

AI Systems: Hybrid Retrieval, Re-ranking, Neo4j, Async AI Serving

Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, HTML, CSS

Certifications: GCP Professional MLOps Engineer, AWS Certified Developer Associate